THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 8 — MATHS CHAPTER TEST SET B

Chapter 11: **Exploring Some Geometric Themes** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Draw neat figures wherever required.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. In the Sierpinski triangle, the number of small shaded triangles after stage 2 is: 1

(a) 12 **(b)** 8 **(c)** 18 **(d)** 9

2. In the Sierpinski triangle, the number of small shaded triangles after stage 6 is: 1

(a) 728 **(b)** 1458 **(c)** 729 **(d)** 732

3. The number of edges of a cuboid is: 1

(a) 15 **(b)** 13 **(c)** 12 **(d)** 11

4. The number of edges of a square pyramid is: 1

(a) 11 **(b)** 9 **(c)** 8 **(d)** 7

5. The number of edges of a pentagonal prism is: 1

(a) 15 **(b)** 16 **(c)** 14 **(d)** 18

6. A solid has 8 faces and 12 vertices. Using $F + V - E = 2$, the number of edges is: 1

(a) 22 **(b)** 18 **(c)** 16 **(d)** 20

7. A two-dimensional pattern that can be folded to make a solid is its: 1

(a) section **(b)** projection **(c)** net **(d)** shadow

8. The faces of a prism that are not bases are called: 1

(a) lateral faces **(b)** vertices **(c)** edges **(d)** nets

9. Which is not a polyhedron? 1

(a) prism **(b)** pyramid **(c)** cylinder **(d)** cube

10. Which solid has all faces as congruent equilateral triangles? 1

(a) square pyramid **(b)** regular tetrahedron
(c) cube **(d)** prism

SECTION — B (Very Short Answer Type) 4 × 2 = 8

Answer in about 10–15 words each.

11. Verify Euler's formula for a triangular prism. 2

12. What is the top view of a cone placed on its base? 2

13. Name the solid whose net has two circles and one rectangle. 2

14. How many faces does a square pyramid have? Name their shapes. 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 30–40 words each.

15. Find the faces, edges and vertices of a hexagonal prism and verify Euler's formula. 3

16. Find the faces, edges and vertices of a octagonal pyramid and verify Euler's formula. 3

17. Find the faces, edges and vertices of a hexagonal pyramid and verify Euler's formula. 3

OR

Find the faces, edges and vertices of a pentagonal prism and verify Euler's formula. 3

18. Find the faces, edges and vertices of a triangular pyramid and verify Euler's formula. 3

OR

Find the faces, edges and vertices of a octagonal prism and verify Euler's formula. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 80–100 words each.

19. (i) Draw the top, front and side views of a cylinder lying on its curved surface. 5
(ii) Name each view.

OR

Case study: A cone-shaped party hat is made from a sector of a circle. 5
(i) Name the solid.
(ii) Number of faces and edges.
(iii) What is the net?

20. (i) How many cubes of edge 1 cm fit in a cuboid of 5 × 4 × 3 cm? 5
(ii) Count the number of cubes on the surface.

OR

(i) What is the cross section of a cube cut diagonally through two opposite edges? 5
(ii) Of a cone cut parallel to the base?

END OF QUESTION PAPER — Total: 40 Marks

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